

◆ 16.2 Unified Core Dynamics

16.2.1 Master Equation

$$\Psi_{k+1} = \Psi_k + dt, F(\Psi_k, L_k, P_k, u_k) + \sigma \sqrt{dt}, \xi_k$$

where:

- $\xi_k \sim \mathcal{N}(0, I)$ is a standard Gaussian noise
 - $dt > 0$ is the time step, chosen sufficiently small for numerical stability
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16.2.2 Unified Force Definition

$$\begin{aligned} F(\Psi_k, L_k, P_k, u_k) = & \\ & -\nabla \mathcal{E}(\Psi_k) \\ & - L_k \Psi_k \\ & + \alpha(PAC_k) \nabla \ln P_k \\ & + B(u_k) \end{aligned}$$

Each term represents:

- $-\nabla \mathcal{E}(\Psi_k)$: energy descent (stabilizing term)
- $-L_k \Psi_k$: network coupling effect
- $\alpha(PAC_k) \nabla \ln P_k$: probabilistic

learning / adaptation

- $B(u_k)$: control input injection

16.2.3 Time-Step Stability Condition

$$\Delta t < \frac{1}{L_{\text{tot}}}$$

where:

$$L_{\text{tot}} = L_E + \|\Lambda\| + \alpha_{\max} C + L_B$$

ensuring:

absence of numerical divergence in the discrete-time system.

16.2.4 Consistency of Dynamics

All components of F depend on the same state Ψ_k , ensuring:

no temporal lag and no structural inconsistency in the system evolution.

16.2.5 Global Lipschitz Property

$$\|F(x) - F(y)\| \leq L_{\text{tot}} \|x - y\|$$

This guarantees:

- well-posedness of the dynamics
- existence and uniqueness (linked to Section 16.1)

16.2.6 Noise Model

$$\sigma \sqrt{dt}, \xi_k$$

with:

- Gaussian distribution
- independent increments

ensuring realistic stochastic behavior.

16.2.7 Control Injection

$$F \supset B(u_k)$$

Thus, the system becomes:

a closed-loop controlled stochastic system

16.2.8 Equilibrium Condition

$$F(\Psi^*, L^*, P^*, u^*) = 0$$

which defines a stationary point:

Ψ^* is an equilibrium state

16.2.9 Link to Mathematical Foundation

This formulation relies on assumptions established in Section 16.1:

- compact domain Ω
- Lipschitz continuity
- convex energy structure

ensuring mathematical validity of the system dynamics.

Closure Statement (Part II)

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\boxed{
\text{The system defines a unified stochastic
closed-loop dynamic engine.}
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**SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
DOI: 10.5281/zenodo.19521219"**